

RESEARCH ARTICLE

Path-Aware Knowledge Injection for Fine-Grained Emotion Recognition in Mental Health Counseling

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ABSTRACT Fine-grained, multi-label emotion recognition in textual dialogues, particularly in contexts like mental health counseling, presents significant challenges due to the complexity and concurrency of human emotions. Standard pre-trained language models, while powerful, often struggle to capture the underlying reasoning and emotional interdependencies that are not explicitly stated in the text. To address this, we propose PK-GAT, a novel framework for Path-aware Knowledge-injected Graph Attention Networks. Our approach moves beyond traditional knowledge fusion by actively reasoning over a commonsense knowledge graph (i.e., SenticNet). It identifies, encodes, and ranks multi-hop semantic paths that connect textual concepts to emotion labels, thereby injecting rich, inferential logic into the model. This path-aware mechanism is integrated with a Graph Attention Network (GAT) that dynamically models the relationships between emotion labels, allowing them to mutually refine one another. We conduct extensive experiments on the GoeMotions and SemEval-2018 datasets. Our results demonstrate that PK-GAT achieves new state-of-the-art performance, attaining a Macro-F1 score of 58.72% and a Micro-F1 of 69.85% on GoeMotions, outperforming strong baselines and recent models. Additionally, through detailed ablation and case studies, we validate the significant contributions of both the path-based reasoning and the dynamic label graph modeling, offering a robust and effective solution for nuanced emotion recognition.

INDEX TERMS Mental health counseling, graph attention network, knowledge graph, multi-label emotion recognition.

I. INTRODUCTION

Emotion recognition from text has grown into a central task in natural language processing (NLP), particularly within the field of mental health counseling, where understanding clients' nuanced emotional states is essential for effective support [1]. The ability to accurately decode fine-grained emotions embedded in everyday narratives enables downstream applications such as automated psychological screening [2], sentiment-aware dialogue agents [3], and real-time mental health interventions [4].

Early research in emotion recognition relied heavily on traditional machine learning approaches that depended on

careful manual feature engineering. Methods like Support Vector Machine (SVM) [5], Logistic Regression (LR) [6], and Random Forest (RF) [7], paired with features such as N-grams [8] and emotion lexicons [9], achieved moderate success but struggled to capture context and subtlety, especially when confronted with colloquial or complex expressions typical of counseling dialogues. In brief, these approaches often fell short in representing the multi-label, overlapping, and context-dependent nature of emotional experience.

The landscape began to shift with the adoption of deep learning, where Convolutional Neural Network (CNN) [10] and Recurrent Neural Network (RNN) [11]—particularly Long Short Term Memory (LSTM) [12] and Gated Recurrent Unit (GRU) [13]—learned distributed text representations that captured both local patterns and sequential

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dependencies. Attention mechanisms further improved performance, enabling models to weigh the importance of words or phrases dynamically [14]. More recently, large-scale pre-trained language models (PLMs) based on the Transformer architecture, such as BERT [15], RoBERTa [16], and DeBERTa [17], have set new performance benchmarks. Their ability to encode deep semantic information from text has made them a popular choice for fine-tuning in emotion recognition tasks.

Despite these advances, several persistent challenges remain. First, emotions are rarely expressed in isolation; they interact and overlap, requiring models to handle multi-label scenarios and learn the complex relationships among emotional states. Second, real-world data, especially in the mental health domain, is often imbalanced, with some emotions vastly underrepresented, making it difficult for models to detect rare but critical signals. Finally, while PLMs offer powerful text encodings, they are limited by the information present in the training data and may fail to make the kind of commonsense or causal inferences that humans rely on when interpreting nuanced emotion.

Recent research has tackled these gaps in several ways. Some approaches incorporate external resources—such as knowledge graphs [18] or emotion hierarchies [19]—to provide structured semantic information and improve generalization. Others have introduced novel loss functions like Asymmetric Loss (ASL) [20] to directly address class imbalance, or have adopted auxiliary tasks that encourage richer label semantics and inter-emotion reasoning [21]. However, many existing frameworks still treat external knowledge in a static, shallow manner or model emotion dependencies only at the label level, missing out on the deeper, multi-step reasoning that underpins real psychological interpretation.

This paper aims to address these gaps by introducing a path-aware knowledge injection framework for fine-grained emotion recognition in mental health counseling texts. Specifically, our method fuses the deep semantic power of advanced PLMs with structured emotional commonsense from an external knowledge graph, explicitly modeling dynamic, multi-step paths that connect concepts in user narratives to target emotions. The framework further leverages a dynamic graph attention mechanism to capture inter-emotion dependencies and incorporates an optimized loss tailored for the imbalanced, multi-label nature of real-world data.

The main contributions of this work are as follows:

- 1) We propose a novel path-aware knowledge injection module, which dynamically encodes multi-step semantic reasoning paths between textual concepts and emotional labels using an external commonsense knowledge graph, allowing for deeper and more contextualized emotion inference.
- 2) We integrate this knowledge injection with a graph attention network (GAT) that explicitly models the contextual dependencies and interactions among emotion labels,

enhancing the model's capacity to handle concurrent and correlated emotions.

- 3) We apply an asymmetric loss function specifically adapted for multi-label, imbalanced emotion data, substantially improving recognition of rare but clinically relevant emotions.

- 4) Comprehensive experiments on both narrative-rich (GoeMotions) [22] and social-media (SemEval-2018 Task 1) datasets [23] show that our framework outperforms strong baselines, establishing new state-of-the-art results and offering qualitative insights into its reasoning process through real-case analyses.

II. RELATED WORKS

The task of emotion recognition from text has evolved significantly, transitioning from traditional statistical methods to sophisticated deep learning architectures. Early approaches in this domain were firmly rooted in machine learning, where the primary workflow involved manual feature engineering followed by classifier training. Researchers extensively explored algorithms like SVM, LR, and various ensemble methods such as RFs. For instance, studies by Parvin et al. [24] and Kuppachi et al. [25] consistently identified LR and SVM as top performers on benchmark datasets for fine-grained emotion classification, showcasing their robustness in handling multi-class scenarios. The efficacy of these models, however, was heavily reliant on the quality of hand-crafted features. These features often included lexical cues like n-grams, TF-IDF scores [26], and specialized emotion lexicons such as WordNet-Affect [27] and EmoLex [28]. While techniques like Pointwise Mutual Information (PMI) [29] were introduced to mitigate the sparsity of term-feature matrices, the entire process remained labor-intensive and struggled to capture the deeper contextual nuances of language.

The advent of deep learning marked a significant shift, offering a new path to automatically learn hierarchical feature representations directly from raw text. Early deep learning efforts in emotion recognition were dominated by CNNs and RNNs. Hassan et al. [30] first demonstrated the power of CNNs for sentence classification, a model favored for its computational efficiency and ability to capture local n-gram patterns. Concurrently, RNN variants like LSTM and GRU became standard choices for tasks requiring the modeling of sequential dependencies in longer texts. To further refine these models, researchers began incorporating attention mechanisms, allowing models to dynamically weigh the importance of different words or text segments, as shown by Fan et al. [31] and Chen et al. [32], who respectively applied attention to CNN and GRU architectures to enhance focus on salient emotional cues.

More recently, the landscape has been reshaped by the dominance of large-scale PLMs based on the Transformer architecture. The fine-tuning paradigm, where a model like BERT or its more robust successor, RoBERTa, is adapted

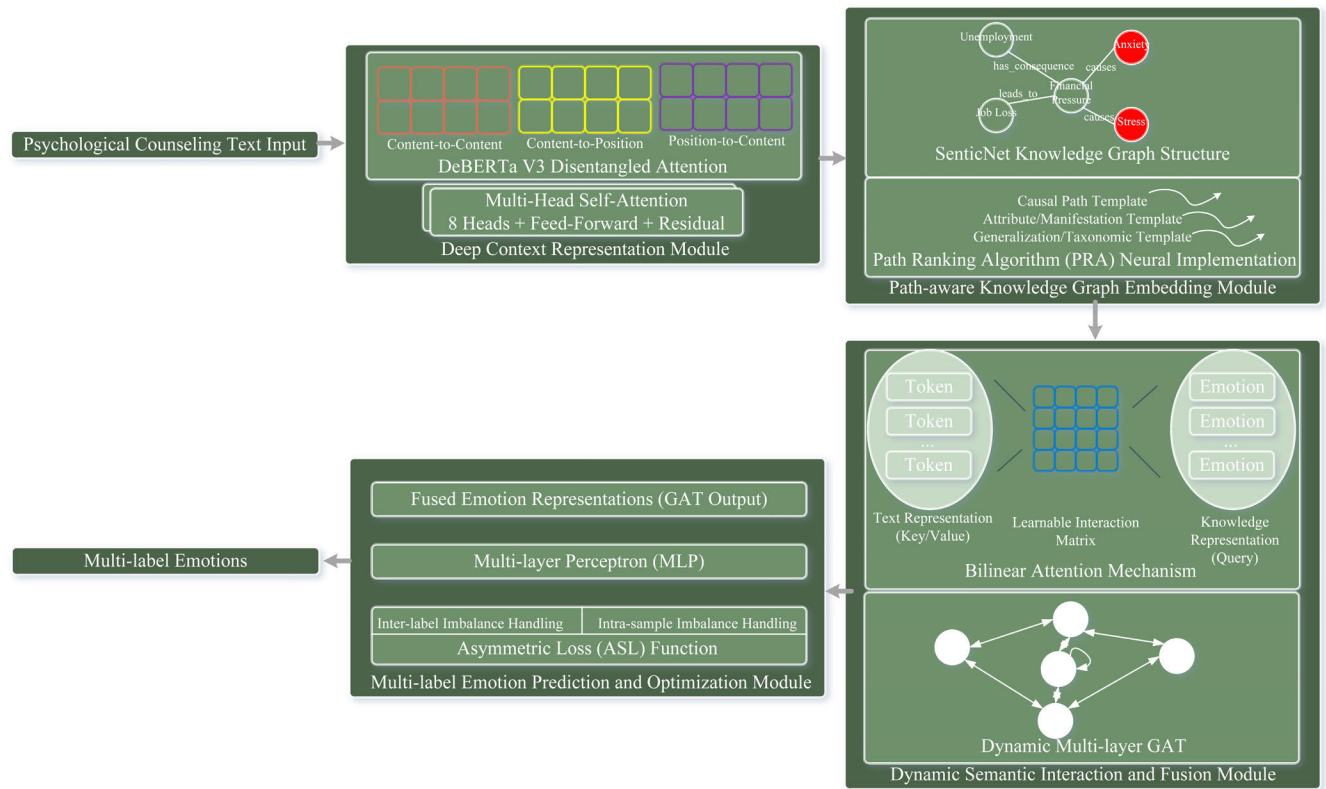


FIGURE 1. Our framework.

for a specific downstream task, has become the de-facto standard. Sitoula et al. [33] provided a clear comparison, affirming that RoBERTa consistently outperforms BERT and CNNs on the GoEmotions dataset. This raw power of PLMs forms the foundation of many contemporary approaches. However, researchers quickly identified that simply fine-tuning a PLM is often insufficient for the complexities of fine-grained, multi-label emotion recognition. Consequently, a new wave of research has focused on augmenting these powerful PLMs with additional mechanisms. For example, López-López et al. [34] explored domain-specific fine-tuning with EmoRoBERTa for educational dialogues, while Ameer et al. [35] and Labib et al. [36] integrated sophisticated multi-head attention and ensemble techniques into RoBERTa-based models to further boost performance on datasets like SemEval-2018 and GoEmotions.

A parallel and highly influential line of inquiry addresses the critical challenge of modeling inter-emotion correlations. Recognizing that emotions rarely occur in isolation, researchers have moved beyond treating labels as independent. Sui et al. [37] proposed an innovative masked emotion modeling mechanism, akin to masked language modeling, which forces a Transformer to learn these correlations implicitly by recovering masked emotion states. Taking a more explicit approach, Chen et al. [38] introduced HGCN-EC, a model that leverages a pre-defined emotion hierarchy, encoding it with a Graph Convolutional Network to guide the

classification process. Another strand of research, exemplified by Singh et al. [39], uses a multi-task learning framework where predicting emotion definitions serves as an auxiliary task, thereby injecting a form of semantic knowledge about the labels themselves. Our work builds upon these insights but distinguishes itself by rejecting static, pre-defined knowledge structures in favor of a dynamic, path-based reasoning mechanism over a commonsense knowledge graph. Furthermore, tackling the pervasive issue of class imbalance in multi-label datasets has become crucial. Ramakrishnan et al. [40] successfully adapted the ASL function, originally from computer vision, to the NLP domain, demonstrating significant F1-score improvements on imbalanced data by integrating it with BERT. Our framework adopts this advanced loss function, but situates it within a more complex, knowledge-aware architecture, aiming to show that the synergy between a powerful loss function and a sophisticated model yields superior results.

III. THE PROPOSED METHOD

A. OVERVIEW

To accurately identify complex and concurrent emotional states within psychological counseling texts, we propose a deep learning framework. Its central idea integrates the deep semantics of user utterances, structured emotional commonsense knowledge from external knowledge graphs, and

intrinsic relationships within emotional label spaces through multi-level, multi-dimensional fusion. The logical structure of our framework, illustrated in Figure 1, consists of four sequential modules:

Firstly, the Deep Context Representation Module (Section III-B) transforms raw counseling texts into deeply contextualized embeddings tailored specifically for this task. This is achieved using a two-stage architecture involving DeBERTa V3 followed by a dual-layer Transformer encoder. Next, in the Path-aware Knowledge Graph Embedding Module (Section III-C), we leverage SenticNet as an external knowledge source. Here, we innovatively adopt a neural network inspired by the Path Ranking Algorithm (PRA) to encode semantic paths linking textual concepts to emotional labels, producing knowledge representations rich in inferential logic. Then, within the Dynamic Semantic Interaction and Fusion Module (Section III-D), a bilinear attention mechanism aligns text with external knowledge representations deeply, while a GAT dynamically models the mutual influences among emotional labels. Lastly, the Multi-label Emotion Prediction and Optimization Module (Section III-E) decodes these fused features and employs a ASL specifically designed for multi-label scenarios, optimizing the model end-to-end.

B. DEEP CONTEXT REPRESENTATION MODULE

Psychological counseling texts frequently include colloquial, unstructured, and subtly nuanced expressions. To effectively capture high-fidelity semantic information from these original narratives, we designed a two-stage deep contextual representation architecture.

1) FUNDAMENTAL SEMANTIC ENCODING

We employ DeBERTa V3 [41] as the foundational text encoder, which notably utilizes a disentangled attention mechanism. Given an input counseling corpus C , we first tokenize it using DeBERTa's tokenizer, resulting in a sequence of token IDs of length L . This ID sequence is then transformed into an initial embedding matrix via an embedding layer. At each layer l (ranging from 1 to $L_{DeBERTa}$), attention scores are decomposed to enhance interpretability. Specifically, for any pair of tokens i and j within the sequence, their attention score $A_{i,j}^{(l)}$ at layer l is computed based on their hidden state vectors from the preceding layer, $\tilde{h}_i^{(l-1)}$ and $\tilde{h}_j^{(l-1)}$, along with their relative positional embedding vector $\tilde{p}_{i,j}$.

Initially, the hidden states from the previous layer are linearly projected through weight matrices W_Q, W_K, W_V to derive the Query, Key, and Value vectors:

$$\begin{cases} \tilde{q}_i^{(l-1)} = \tilde{h}_i^{(l-1)} W_Q \\ \tilde{k}_j^{(l-1)} = \tilde{h}_j^{(l-1)} W_K \\ \tilde{v}_j^{(l-1)} = \tilde{h}_j^{(l-1)} W_V \end{cases} \quad (1)$$

The attention scores are further separated into interactions involving content and positional embeddings, with positional embeddings undergoing additional projections via matrices

$W_{K,pos}$ and $W_{Q,pos}$:

$$\begin{cases} \tilde{A}_{i,j}^{c2c} = \tilde{q}_i^{(l-1)} (\tilde{k}_j^{(l-1)})^T & (\text{content} - \text{to} - \text{content}) \\ \tilde{A}_{i,j}^{c2p} = \tilde{q}_i^{(l-1)} (\tilde{p}_{i,j} W_{K,pos})^T & (\text{content} - \text{to} - \text{position}) \\ \tilde{A}_{i,j}^{p2c} = (\tilde{p}_{j,i} W_{Q,pos}) (\tilde{k}_j^{(l-1)})^T & (\text{position} - \text{to} - \text{content}) \end{cases} \quad (2)$$

These interaction terms are summed and scaled by a factor $\frac{1}{\sqrt{3d_k}}$, where d_k denotes the dimensionality per attention head, resulting in the final attention score:

$$A_{i,j}^{(l)} = \frac{1}{\sqrt{3d_k}} (\tilde{A}_{i,j}^{c2c} + \tilde{A}_{i,j}^{c2p} + \tilde{A}_{i,j}^{p2c}) \quad (3)$$

After normalization via softmax, these attention scores weight the Value vectors to update each token's hidden state $\tilde{h}_i^{(l)}$. Following this deep interaction across $L_{DeBERTa}$ layers, the output from the last DeBERTa layer constitutes our final deep contextual embedding matrix T . This matrix, with dimensions $L \times d_{model}$, can be succinctly described as:

$$T = DeBERTaV3(C) \quad (4)$$

2) TASK-ORIENTED FEATURE RE-FUSION

To enhance the general semantic features generated by DeBERTa V3, making them more tailored to our specific emotion recognition task, we introduce a lightweight two-layer Transformer encoder. The encoder accepts DeBERTa V3's output matrix T as its input, denoted clearly here as $E^{(0)}$ to maintain mathematical clarity and layer-by-layer logical progression.

Within each Transformer encoder layer (indexed by m from 1 to 2), we begin with a Multi-Head Self-Attention (MHSA) operation. Initially, the input matrix $E^{(m-1)}$ undergoes linear projections into queries, keys, and values. These are partitioned into H distinct heads, each with separate projection matrices $W_{Q,h}, W_{K,h}, W_{V,h}$:

$$\begin{cases} Q_h = E^{(m-1)} W_{Q,h} \\ K_h = E^{(m-1)} W_{K,h} \\ V_h = E^{(m-1)} W_{V,h} \end{cases} \quad (5)$$

Each head captures varied dependencies across distinct representation subspaces, with attention outputs $head_h$ computed as follows:

$$head_h = \text{soft max} \left(\frac{Q_h K_h^T}{\sqrt{d_k/H}} \right) V_h \quad (6)$$

The outputs from all heads are concatenated and combined through an output projection matrix W_O :

$$MHSA(E^{(m-1)}) = \text{Concat}(head_1, head_2, \dots, head_H) W_O \quad (7)$$

Following this, the MHSA output integrates with the original input $E^{(m-1)}$ via a residual connection, subsequently passing through Layer Normalization (LayerNorm). This result then feeds into a Position-wise Feed-Forward Network

(FFN), comprising two linear transformation layers defined by weight matrices and biases:

$$E'^{(m)} = \text{LayerNorm}(E^{(m-1)} + \text{MHSA}(E^{(m-1)})) \quad (8)$$

$$\text{FFN}(E'^{(m)}) = \text{ReLU}(E'^{(m)}W_1 + \tilde{b}_1)W_2 + \tilde{b}_2 \quad (9)$$

Again, the FFN output is merged via a residual connection and LayerNorm to yield the final output matrix of the m -th layer, denoted as $E^{(m)}$:

$$E^{(m)} = \text{LayerNorm}(E'^{(m)} + \text{FFN}(E'^{(m)})) \quad (10)$$

After these two layers, we achieve the final textual representation E_{text} . Thus, the overall process is summarized succinctly as:

$$E_{\text{text}} = \text{TransformerEncoder}_{2\text{-Layer}}(T) \quad (11)$$

This two-stage approach not only refines the deep semantic understanding of each token but also effectively captures the dynamic emotional trajectory throughout the counseling dialogue, laying a robust foundation for subsequent knowledge integration tasks.

C. PATH-AWARE KNOWLEDGE GRAPH EMBEDDING MODULE

Traditional GNN-based knowledge graph embedding methods [42], [43] mainly focus on immediate neighbors of nodes, neglecting the rich logical and inferential information encoded in multi-step semantic paths connecting entities. To address this limitation, we propose a path-aware knowledge injection mechanism inspired by the Path Ranking Algorithm (PRA). Specifically, our method first identifies candidate semantic paths between source and target entities, then learns a ranking function to evaluate the significance of each path, and finally aggregates the important path information. Unlike traditional PRA that relies on manually defined features, our framework employs an end-to-end neural network implementation.

We select SenticNet [44] as our external knowledge base due to several key advantages. SenticNet is concept-centric, capturing abstract ideas effectively, and importantly, it provides a four-dimensional emotional vector—Pleasantness, Attention, Sensitivity, and Aptitude—based on the psychological Hourglass of Emotions model. These dimensions offer detailed insights into emotional valence, arousal, impact, and control, surpassing simple positive/negative categorization and aligning closely with professional psychological assessments. To illustrate SenticNet's structure concretely, we visualize a subgraph around the core emotion “Happiness,” as depicted in Figure 2. Concepts related to happiness are grouped into semantic clusters such as “Positive Experience,” “Cognitive State,” and “Social Emotion.” Nodes are interconnected by edges labeled with semantic relationships (e.g., intensifies, results_from, manifests_as), forming meaningful inferential paths. Each concept node includes its specific four-dimensional emotional vector values

(P, A, S, Ap), quantifying emotional properties. This structured representation enables the extraction of deep semantic paths linking textual concepts to specific emotional labels.

Consequently, for key conceptual entities identified in texts and our targeted emotional labels, we extend beyond mere node embeddings by mining semantic paths (e.g., from “unemployment” to “anxiety” via the path “unemployment” $\rightarrow$ has_consequence $\rightarrow$ “financial pressure” $\rightarrow$ causes $\rightarrow$ “anxiety”). The detailed process unfolds in four main steps:

1) CANDIDATE PATH GENERATION

To link key textual concepts (i.e., source node, c_{src}) to targeted emotional labels (i.e., target node, e_{tgt}), we first extract meaningful candidate paths using a constrained random walk guided by predefined meta-path templates. These templates reflect psychological logic and abstract SenticNet's original relationship types into generalized categories, improving the robustness and efficiency of path search. Specifically, we define three primary meta-path templates:

(1) Causal Path Template: Concept $\rightarrow$ causes $\rightarrow$ Concept/Emotion, instantiated by relationships such as Causes, HasSubevent, Motivates, HasConsequence.

(2) Attribute/Manifestation Path Template: Concept $\rightarrow$ manifests_as $\rightarrow$ Emotion, instantiated via HasProperty, DefinedAs, ObstructedBy relationships.

(3) Generalization/Taxonomic Path Template: Concept $\rightarrow$ is_a $\rightarrow$ Concept, instantiated by IsA, TypeOf, PartOf relationships.

Adhering to these templates ensures logical validity, generating a candidate path set $P(c_{\text{src}}, e_{\text{tgt}}) = \{\text{path}_1, \text{path}_2, \dots, \text{path}_K\}$, where K represents the total number of candidate paths discovered.

2) PATH ENCODING

A semantic path path_k comprises alternating nodes and relationships: $\text{path}_k = (c_0, r_1, c_1, \dots, r_S, c_S)$, where c_0 is the source concept node of the path, namely c_{src} ; c_S is the target emotion label node of the path, namely e_{tgt} ; c_s (for $0 < s < S$) is the intermediate conceptual node that the path passes through; r_s is the relationship (or edge) connecting the node c_{s-1} and c_s . To encode this heterogeneous sequence into a fixed-length vector p_k , we employ a dual-branch embedding strategy integrated via a single-layer Transformer encoder.

First of all, we adopt different embedding methods for different types of elements (i.e., nodes and relationships) in the path. For the relation r_s in the path, we maintain a learnable relation embedding matrix E_R and obtain its vector representation through table lookup operations. For the concept node c_s in the path, we utilize its predefined four-dimensional sentiment vector v_s in SenticNet. Since the original embedding dimensions of nodes and relations are inconsistent, we introduce a linear projection layer Proj_N to map the 4-dimensional node sentiment vectors to the same d_k -dimensional space as the relation embeddings. This step

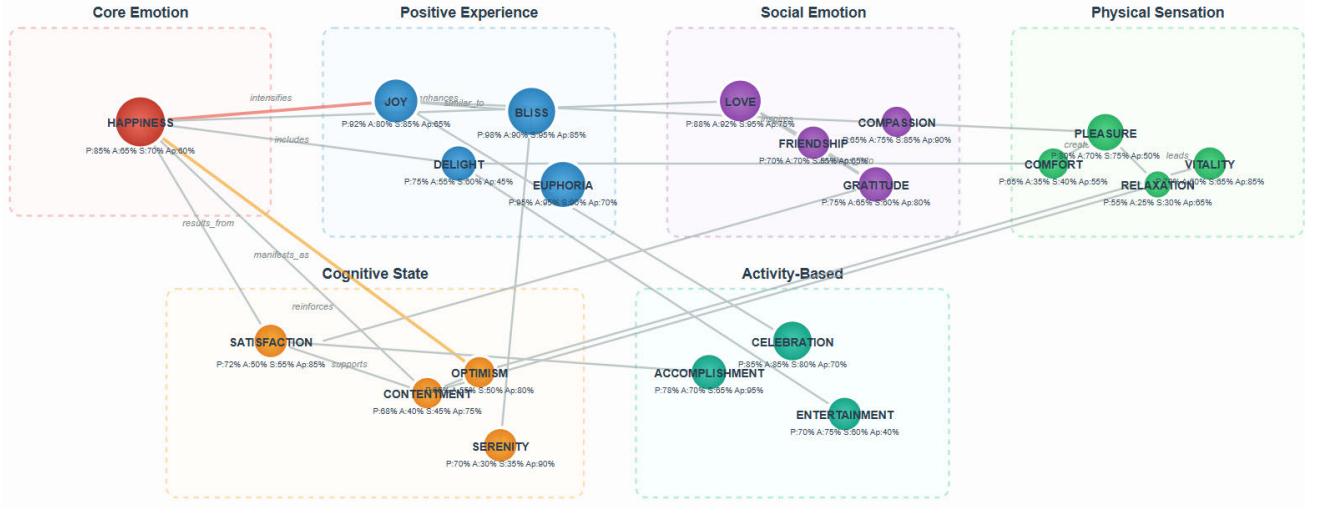


FIGURE 2. The part of the knowledge graph related to the emotion of happiness in SenticNet.

not only achieves dimensional alignment, but also enables the model to learn how to transform the original emotional features into a representation space that is more conducive to the task.

Subsequently, we interleave the projected node embeddings and relation embeddings in the order in which they appear in the path to form a homogeneous vector sequence. We place a learnable [CLS] token embedding at the beginning of this sequence. The final formed input matrix E_{path_k} is sent to the single-layer Transformer encoder.

$$E_{path_k} = [CLS] \oplus Interleave(\{E_R(r_s)\}_{s=1}^S, \{Pr_{ojN}(v_s)\}_{s=0}^S) \quad (12)$$

In which, $\oplus$ and *Interleave* respectively represent the operations of concentration and interleaving. The final output embedding is taken from the Transformer's [CLS] token representation:

$$p_k = TransformerEncoder_{path,1-Layer}(E_{path_k})[CLS] \quad (13)$$

3) PATH IMPORTANCE RANKING

We introduce a two-layer Multi-Layer Perceptron (MLP) as a ranking function to evaluate each candidate path's importance score ρ_k directly from its comprehensive path embedding p_k :

$$\rho_k = Sigmoid(ReLU(p_k W_{mlp,1} + b_{mlp,1}) W_{mlp,2} + b_{mlp,2}) \quad (14)$$

4) PATH INFORMATION AGGREGATION

For each emotional label l_i , we aggregate candidate paths terminating at $e_{tgt} = l_i$, weighted by their importance scores ρ_k . This forms the final path-aware knowledge representation k_i :

$$k_i = \sum_{k \in P(c_{src}, l_i)} \rho_k \cdot p_k \quad (15)$$

Thus, the resulting knowledge representations $E_{knowledge} = \{k_1, k_2, \dots, k_{N_e}\}$ encapsulate dynamic, logically-rich path

aggregations rather than isolated node embeddings. In which, N_e represents the total number of emotion categories.

D. DYNAMIC SEMANTIC INTERACTION AND FUSION MODULE

This module establishes connections between textual representations and path-aware knowledge representations, dynamically modeling the interdependencies among emotional labels.

1) BILINEAR ATTENTION-BASED SEMANTIC INTERACTION

To achieve finer semantic alignment between textual and knowledge representations, we employ a bilinear attention mechanism. Compared to standard dot-product attention, this approach uses a learnable interaction matrix W_{att} , enhancing the model's ability to capture complex, nonlinear cross-modal relationships.

Specifically, let the path-aware knowledge representation k_i of each emotion label (as Query) "examine" the deep context representation matrix E_{text} of the entire text (as the source of Key and Value). And calculate the importance of each lexical element in the text to this emotion label. For the i -th emotion label and the j -th text lexical element, their attention scores $\omega_{i,j}$ are calculated as follows, where e_j is the j -th row of the matrix E_{text} :

$$\omega_{i,j} = \frac{\exp(k_i^T W_{att} e_j)}{\sum_{k=1}^L \exp(k_i^T W_{att} e_k)} \quad (16)$$

Using these computed attention weights $\omega_{i,j}$, we dynamically extract and aggregate contextually tailored textual views for each emotion label l_i . The resulting vector v_i captures the semantic segments within the text most relevant to emotion l_i , effectively forming a weighted textual representation from the specific emotional perspective:

$$v_i = \sum_{j=1}^L \omega_{i,j} e_j \quad (17)$$

2) LABEL RELATIONSHIP MODELING WITH GRAPH ATTENTION NETWORKS

According to psychological theories, emotions do not exist in isolation; they inherently influence one another. To flexibly model this interplay, we replace traditional static Graph Neural Networks (e.g., GGNN) [45] with Graph Attention Networks (GAT). GAT allows each emotion node to dynamically assign varying attention weights to adjacent emotion nodes based on contextual cues, thus capturing context-dependent and asymmetric relationships more effectively.

We initiate GAT with context-aware label representations $V = \{v_1, v_2, \dots, v_{N_e}\}$, where N_e denotes the total number of emotional categories. GAT updates node representations through multiple stacked attention layers. Within each layer, the updated representation of node i aggregates information from neighboring nodes $j \in N_i$, with aggregation weights $\alpha_{i,j}$ computed via a dynamic self-attention mechanism:

$$\alpha_{i,j} = \frac{\exp(\text{LeakReLU}(a^T [Wv_i || Wv_j]))}{\sum_{k \in N_i} \exp(\text{LeakReLU}(a^T [Wv_i || Wv_k]))} \quad (18)$$

$$g_i = \sigma \left(\sum_{j \in N_i} \alpha_{i,j} Wv_j \right) \quad (19)$$

In which, W represents a learnable linear transformation matrix, and a is the attention mechanism's weight vector. After several iterations through multiple GAT layers, we obtain deeply integrated emotional representations $G_{out} = \{g_1, g_2, \dots, g_{N_e}\}$ that encapsulate the dynamically modeled label relationships.

E. MULTI-LABEL EMOTION PREDICTION AND OPTIMIZATION MODULE

With the refined emotional representations G_{out} obtained via the GAT, we construct a lightweight MLP for prediction. A Sigmoid activation function is used to produce prediction probabilities for each emotional category:

$$\hat{y} = \text{Sigmoid}(\text{Linear}_2(\text{ReLU}(\text{Linear}_1(G_{out})))) \quad (20)$$

To optimize the model parameters, we employ an Asymmetric Loss (ASL) function [20], specifically designed for multi-label classification scenarios. ASL effectively addresses dual class imbalances commonly found in emotional datasets:

1) Inter-label imbalance: There is a substantial disparity in the frequency of different emotional labels. For instance, labels such as "joy" or "admiration" appear far more frequently than "grief" or "remorse" within datasets like GoEmotions [22] (as show in Figure 3). This imbalance typically causes models to favor frequent labels during training, neglecting rarer but critically important negative emotions, ultimately hindering overall performance metrics like Macro-F1.

2) Intra-sample imbalance: Within multi-label scenarios, the number of positive labels for any given sample is significantly fewer than negative labels. A typical GoEmotions instance might possess only 2-3 positive emotional labels but

more than 20 negative ones. Traditional Binary Cross Entropy (BCE) loss equally weighs all labels, thus the gradients from numerous easily identified negative labels overwhelm the essential learning signals from the sparse positive labels.

In addition, while subsequent Focal Loss [46] alleviates issues related to simple versus difficult samples via the focusing parameter γ , it typically handles positive and negative samples symmetrically, failing to fully address intra-sample imbalances. In contrast, ASL addresses these dual imbalances effectively by introducing separate focusing parameters γ_+ and γ_- . A smaller γ_+ can be set for rare positive samples to avoid overly suppressing their loss contributions, while a larger γ_- significantly reduces the weights of numerous trivial negative samples. Moreover, ASL employs a probability shifting mechanism to entirely remove the loss contributions of extremely low-probability negative samples (e.g., $p < 0.05$). This approach directs the model's focus toward challenging negative samples (predicted with higher probabilities) and all positive samples. The mathematical form of ASL is as follows:

$$J_{ASL} = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^{N_e} L_{i,j} \quad (21)$$

In which, M is the batch size and N_e is the total number of emotion categories. Each label-specific loss $L_{i,j}$ is defined as:

$$L_{i,j} = y_{i,j}(1 - p_{i,j})^{\gamma_+} \log(p_{i,j}) + (1 - y_{i,j})p_{i,j}^{\gamma_-} \log(1 - p_{i,j}) \quad (22)$$

where $p_{i,j}$ denotes the predicted probability for the j -th label of the i -th sample. By minimizing ASL, our framework is effectively guided to learn and accurately decode complex emotional states from user narratives.

IV. EXPERIMENT

A. DATASET AND PREPROCESSING

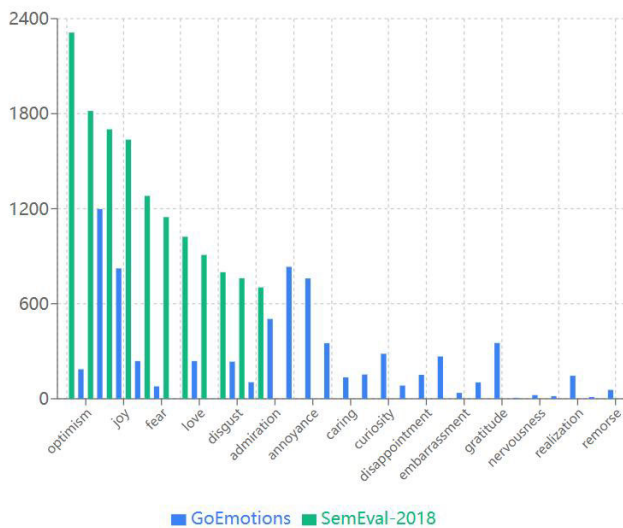
To thoroughly evaluate the effectiveness of the proposed path-aware knowledge injection and dynamic graph attention network framework, we selected representative and widely recognized public datasets. Our selection adhered to three key principles: (1) multi-label characteristics to assess the model's ability in handling concurrent emotions and to verify the efficacy of the ASL; (2) contextual richness and textual diversity to evaluate the model's performance on both narrative-rich, long-form texts and brief, informal social media texts; (3) coverage of real-world concepts to ensure adequate practical scenarios for the path-aware knowledge graph embedding module. Consequently, we chose GoeMotions [22] and SemEval-2018 Task 1: Affect in Tweets (E-c) [23] datasets as our primary experimental platforms. Detailed statistics are provided in Table 1, with emotional distribution visualized in Figure 3.

1) DATASETS

The GoeMotions dataset serves as our primary experimental and evaluation platform. Originating from Reddit, GoeMotions is a large-scale, fine-grained multi-label emotional

TABLE 1. Dataset feature comparison: GoeMotions vs SemEval-2018 (E-C).

Feature	GoEmotions	SemEval-2018 Task 1 (E-c)
Total Sample Size	58,009 samples	10,983 samples
Emotion Categories	27 emotions + neutral (28 total)	11 emotions
Emotion Labels	Admiration, Amusement, Anger, Annoyance, Approval, Caring, Confusion, Curiosity, Desire, Disappointment, Disapproval, Disgust, Embarrassment, Excitement, Fear, Gratitude, Grief, Joy, Love, Nervousness, Optimism, Pride, Realization, Relief, Remorse, Sadness, Surprise + Neutral	Anger, Anticipation, Disgust, Fear, Joy, Love, Optimism, Pessimism, Sadness, Surprise, Trust
Data Source	Reddit comments	Twitter tweets
Platform Characteristics	Long-form discussions, threaded conversations	Short-form microblogging, 280 character limit
Annotation Method	Multi-label classification	Multi-label classification + intensity scoring
Intensity Scoring	Binary (present/absent)	4-point scale (0-3) for intensity
Average Text Length	~31 tokens	~23 tokens
Text Length Range	Variable (Reddit comment format)	Limited by Twitter character constraints
Training/Dev/Test Split	43,410 / 5,426 / 5,427	6,838 / 886 / 3,259


FIGURE 3. Emotion categories distribution comparison (Note: the neutral category accounts for approximately 47% of the entire GoEmotions dataset. To present the data without bias, this category is omitted in the figure).

classification dataset, primarily collected from personal experience and emotional expression subreddits like *r/self* and *r/relationships*. This thematic and narrative similarity aligns closely with our target application scenario of psychological counseling. Its features match the core innovations of our framework, containing 27 non-exclusive fine-grained emotional categories (e.g., admiration, annoyance, grief, nervousness, remorse) and a neutral category, making it ideal for assessing our model’s multi-label capabilities and the ASL function. Additionally, GoeMotions texts usually involve long narratives rich in context, thus providing ample opportunities for our DeBERTa V3 + Transformer-based deep contextual representation module. Critically, these authentic narratives are replete with real-life events and concrete concepts, enriching the inference source concepts for our SenticNet-based knowledge graph embedding module.

Additionally, we utilized the SemEval-2018 Task 1: Affect in Tweets (E-c) dataset as a complementary evaluation to assess our framework’s generalization and robustness. This authoritative dataset, designed for social media sentiment analysis tasks, differs from GoeMotions by containing numerous short, noisy, and colloquial tweets labeled across 11 basic emotions (e.g., anger, joy, sadness, fear). Testing on this dataset evaluates the stability and effectiveness of our model in dealing with sparse, context-limited short texts. Furthermore, as a widely-used academic benchmark, it facilitates direct and unbiased comparisons with contemporary state-of-the-art works.

2) DATA PREPROCESSING

To eliminate noise and standardize texts for model input, we developed customized preprocessing workflows for each dataset. We adhered to official data splits to ensure comparability of results, as shown in Table 1.

For the relatively standardized GoeMotions dataset, our preprocessing included:

- 1) Text cleaning: URLs, email addresses, and Reddit-specific formatting were removed.
- 2) Lowercasing: Texts were converted entirely to lowercase to standardize vocabulary representation.
- 3) Punctuation and special character handling: Emotionally expressive punctuation (e.g., periods, question marks, exclamation marks) was retained, while unnecessary special characters and excessive whitespace were removed.

For the SemEval-2018 Task 1 dataset, with its inherently informal and noisy tweets, preprocessing steps were more comprehensive:

- 1) Standardization of special entities: User mentions (e.g., @someuser) were replaced with a uniform placeholder < USER >, hashtags (e.g., #topic) had their “#” removed but retained the content as normal text.

2) URL removal and lowercasing: All URLs were removed, and texts were lowercased similarly to the GoeMotions processing.

3) Emoji handling: Given the emotional significance of emojis in social media, we used Python-emoji toolkit to translate emojis into descriptive text (e.g., “:)” becomes “smile”), transforming visual cues into textual signals interpretable by the model.

4) De-duplication and normalization: We normalized repeated characters (e.g., “soooo good” became “so good”) to reduce textual noise.

After preprocessing, texts were tokenized using DeBERTa V3’s tokenizer, preparing them suitably for subsequent model processing.

B. EXPERIMENTAL SETUP

All experiments were conducted on servers equipped with NVIDIA Tesla V100 GPUs (32GB memory). The software environment comprised Ubuntu 20.04 LTS, Python 3.8, PyTorch 1.12.1, and Transformers 4.21.1.

In the deep contextual representation module, we initialized the base encoder with pre-trained weights and tokenizer from microsoft/deberta-v3-base, resulting in an output dimension d_{model} of 768. The subsequent two-layer Transformer encoder shared the same hidden dimension of 768, utilized 8 attention heads ($H = 8$), and had a feed-forward network with an intermediate dimension of 3072. Within the path-aware knowledge graph embedding module, both the relation embeddings and the projected node emotion embeddings were set to a dimension d_k of 128. The single-layer Transformer encoder for path encoding employed 4 attention heads. Additionally, the two-layer MLP used for path ranking featured a hidden layer size of 64. In the dynamic semantic interaction and fusion module, the learnable interaction matrix W_{att} for bilinear attention was of size 768×128 . For the GAT employed in label relationship modeling, we stacked two layers, each comprising 4 attention heads with an output dimension of 128 per head.

We trained the model end-to-end using the AdamW optimizer [47], with an initial learning rate determined via a hierarchical strategy. Specifically, we applied a smaller learning rate ($1E-5$) to the lower-level DeBERTa V3 encoder to retain its pre-trained knowledge effectively. A higher learning rate ($1E-4$) was adopted for newly introduced parameters in the upper layers. To enhance training stability, we implemented a linear warm-up scheduler with warm-up steps comprising 10% of the total training steps. A dropout rate of 0.1 was consistently applied to all fully connected layers and attention weights to prevent overfitting. Training was conducted with a batch size (i.e., M in the eq.(21)) of 32. For ASL, a grid search on the validation set determined the optimal focusing parameters as $\gamma_+ = 0$ and $\gamma_- = 2$. Early stopping was enforced when the macro-averaged F1 score on the validation set ceased to improve, using a patience of 5 epochs.

Considering the characteristics of multi-label classification and prevalent class imbalance, we report both macro-averaged and micro-averaged F1 scores. The micro-average treats predictions for each sample equally, thus reflecting the overall performance on frequent categories more effectively. Conversely, the macro-average computes individual F1 scores per class before averaging, thereby equally emphasizing each category and highlighting performance on rare classes, crucial for evaluating class imbalance handling. Additionally, we include the Hamming Loss metric [48], which measures the proportion of incorrectly predicted labels, offering complementary insights into the model’s classification errors.

C. EXPERIMENTAL SETUP

In this study, all training, testing, and method comparisons were conducted on a desktop computer running Ubuntu 18.04, equipped with an AMD Ryzen 3-3100X CPU, an AMD Radeon RX 6700 XT GPU, and 32GB of memory.

For the proposed MAKEstimator, the input consists of RGB cartoon images of size 256×256 , and the output consists of 68 heatmaps of size 112×112 . We train the keypoint estimation network for 100 epochs using the Adam optimizer with a batch size of 32 and a learning rate of $1e-3$. The learning rate is decayed by a factor of 0.1 at the 40-th and 80-th epochs. For data augmentation, all faces are cropped to the center and resized to 256×256 , with a rotation of ± 15 degrees. For the loss function, the coordinates of the keypoints are normalized to the range $[-0.5, 0.5]$, and the regularization parameters are set to $\lambda_1 = 1e-5$ and $\lambda_2 = 1e-2$.

For the proposed FRDeformer, we use the RMSProp (Root Mean Square Propagation) optimizer with a batch size of 32 and train the generation network for 1,000 epochs. The learning rate for the first 500 epochs is $1e-3$, and for the last 500 epochs, the learning rate is $1e-4$. In our method, the number of sample points is $N = 16,384$, and the Fourier representation projection parameters $a_i = 1$, with b_i being a vector sampled from a standard normal distribution of dimension (3, 64). The SIREN consists of five layers, where the first layer projects the position encoding to 128 dimensions, the three intermediate hidden layers have 128-dimensional feature maps, and the final output layer has 3-dimensional representations. The unified fully connected layers in the unified layer hyper-network have representation dimensions of 128 and 256. The loss function regularization parameters are set to $\lambda_3 = 1e-3$ and $\lambda_4 = 1e-6$.

D. ABLATION EXPERIMENTS

We performed comprehensive ablation studies on the more challenging and contextually rich GoeMotions dataset to evaluate the contributions and necessity of each core component within the proposed framework. These experiments aimed to quantify the performance gain from each technical innovation and analyze the model’s sensitivity to critical hyperparameters. To ensure fairness, we used the same

experimental settings and hardware/software environment as in our main experiments.

1) ARCHITECTURAL ABLATION

We systematically removed or substituted key components of the proposed model to clearly quantify their individual contributions, as shown in Table 2.

As shown in Table 2, Each component plays a crucial role in the final performance. The largest drop, by 2.84%, occurs in variant M4 when replacing ASL with standard BCE loss, underscoring ASL's effectiveness in addressing severe class imbalance, especially for recognizing rare emotions.

Removing the Path-aware Knowledge Injection (PKI) module (i.e., variant M1) leads to a notable decline of 1.81% in Macro-F1, strongly supporting our hypothesis that structured, multi-step reasoning from external knowledge graphs significantly enhances the textual encoder's ability to understand emotional contexts.

The absence of GAT (i.e., variant M2) results in a Macro-F1 reduction of 1.18%, highlighting the importance of explicitly modeling interdependencies between emotional labels. GAT's dynamic attention mechanism effectively captures co-occurrence patterns, thus refining predictions.

Replacing bilinear attention with simpler dot-product attention (i.e., variant M3) still causes a measurable decrease of 0.69%. This indicates that bilinear attention, offering additional modeling capacity for complex nonlinear alignments between text and knowledge, provides valuable benefits beyond basic attention mechanisms.

2) HYPERPARAMETER SENSITIVITY

To better understand our model's robustness, we analyzed several critical hyperparameters and design choices.

a: NUMBER OF GAT LAYERS

The GAT's depth dictates how extensively information propagates across the label graph. We tested varying GAT layers (1 to 4, the results as shown in Figure 4), with performance peaking at two layers (58.72% Macro-F1). One-layer GAT was insufficient (58.15%), likely due to limited modeling of complex indirect relationships. However, increasing to three or four layers slightly degraded performance (58.41% and 58.26%, respectively), possibly due to over-smoothing, a common issue in deeper graph networks where node representations lose discriminative power. Hence, a two-layer GAT optimally balances representational power and stability.

b: DEPTH OF PATH ENCODER

We examined the effect of varying Transformer encoder depths (1 to 3 layers) for path encoding, the results as shown in Figure 5. A single-layer encoder yielded the highest performance (58.72% Macro-F1). Performance remained stable at two layers (58.69%) but declined slightly at three layers (58.55%). This aligns with our assumption that semantic paths, characterized by structured and limited vocabulary, are effectively captured by a strong single-layer attention

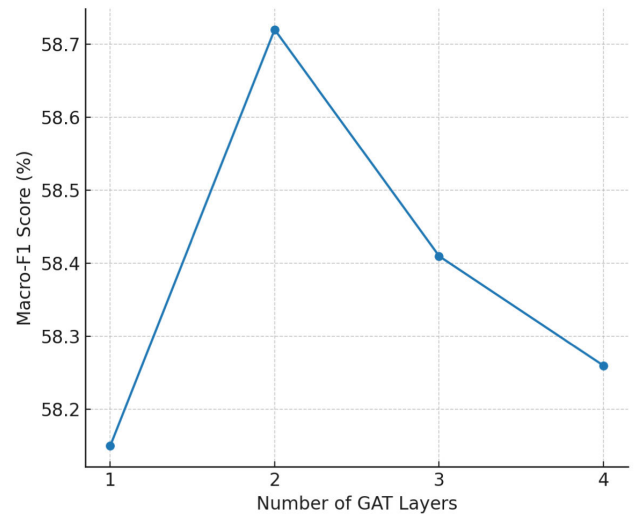


FIGURE 4. The influence of the number of GAT layers on the results.

mechanism. Additional layers provided negligible improvements while increasing complexity and computational load, validating our choice of a single-layer encoder.

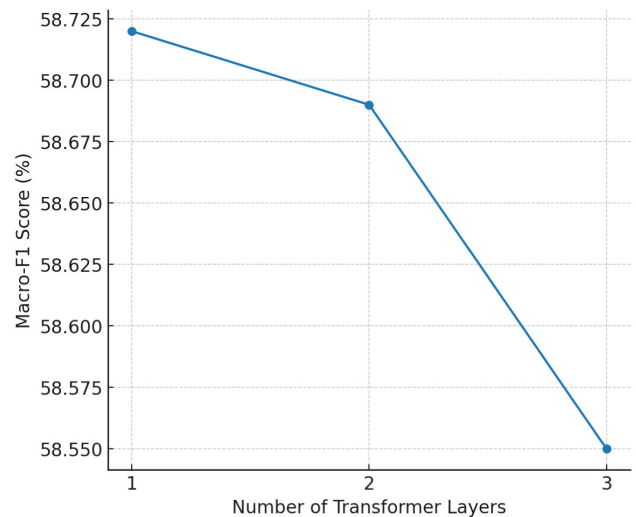


FIGURE 5. The influence of the number of path encoder layers on the results.

c: KNOWLEDGE INJECTION STRATEGIES COMPARISON

We evaluated the proposed path-aware knowledge injection strategy against two baseline methods:

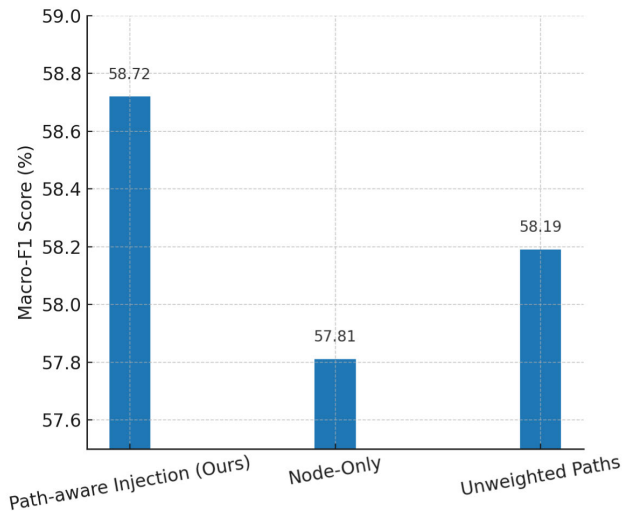
(1) Node-Only: Directly embedding SenticNet concept nodes without path information.

(2) Unweighted Paths: Averaging all path representations without learned weights.

The results from Figure 6 shows our approach notably outperforming both baselines. Compared to Node-Only, it improves Macro-F1 by nearly 1%, emphasizing the value of explicitly modeling multi-step reasoning paths. Compared

TABLE 2. Ablation study of model components on GoeMotions dataset (Macro-F1 and Micro-F1 scores).

Model Variant	Description	Macro-F1	Δ	Micro-F1	Δ
M0 (Full Model)	Our complete framework	58.72	-	69.85	-
M1 (w/o PKI)	Without Path-aware Knowledge Injection module	56.91	-1.81	68.32	-1.53
M2 (w/o GAT)	Replacing GAT with simple MLP classifier	57.54	-1.18	68.99	-0.86
M3 (w/o Bilinear Attn)	Replacing bilinear attention with dot-product attention	58.03	-0.69	69.31	-0.54
M4 (w/o ASL)	Replacing ASL with standard BCE loss	55.88	-2.84	69.05	-0.80

**FIGURE 6.** Performance comparison of different knowledge injection strategies.

to Unweighted Paths, the 0.53% improvement highlights the efficacy of our path-ranking module, effectively weighting more relevant and reliable evidence chains over potentially noisy paths.

E. CONTRAST EXPERIMENTS

This section provides detailed experimental results for our proposed Path-aware Knowledge Injection with Graph Attention Networks (PK-GAT) framework. We benchmarked PK-GAT against diverse and state-of-the-art (SOTA) models on two prominent datasets: GoeMotions and SemEval-2018 Task 1 (E-c). Our analysis highlights quantitative performance improvements and offers qualitative insights into why PK-GAT excels at the challenging task of fine-grained, multi-label emotion recognition.

1) PERFORMANCE COMPARISON WITH STATE-OF-THE-ART MODELS

We selected several representative recent SOTA models spanning fine-tuning advanced pretrained language models (PLMs) to incorporating various external knowledge types. This diverse comparison places our work in a broad academic context, highlighting the unique strengths of our path-aware

knowledge injection and dynamic label graph modeling approaches.

a: PERFORMANCE ON THE GoeMotions DATASET

The GoeMotions dataset, featuring 27 fine-grained emotion categories and rich narrative contexts, serves as a rigorous benchmark.

As shown in Table 3, PK-GAT achieves new state-of-the-art results across all metrics on GoeMotions. It excels notably in Macro-F1, which is sensitive to rare classes, demonstrating its robustness in addressing class imbalance. PK-GAT also attains the lowest Hamming loss (0.098), underscoring its accuracy across all possible label predictions.

Compared to strong pretrained models like RoBERTa [33] and EmoRoBERTa [34], PK-GAT surpasses them by over 4% in Macro-F1, indicating significant improvements not achievable through purely textual encoders. This validates our core hypothesis that integrating structured external knowledge alongside PLMs substantially enhances performance.

Additionally, PK-GAT outperforms other models that incorporate external knowledge. Singh et al. [39] and Chen et al. [38], although effective, trail our model by 1.62% and 1.22% in Macro-F1, respectively. Unlike their approaches that utilize static, high-level knowledge, our method dynamically exploits fine-grained, multi-step semantic paths closely resembling human logical reasoning.

b: PERFORMANCE ON SemEval-2018 TASK 1 (E-C) DATASET

To evaluate generalizability and robustness, we also tested PK-GAT on the SemEval-2018 dataset characterized by short, informal texts.

From Table 4, in the challenging SemEval-2018 benchmark, PK-GAT again demonstrates superior performance, underscoring its effectiveness even with limited contextual information. This result emphasizes the importance of external knowledge integration in processing short, ambiguous texts.

Our model notably outperforms RoBERTa-MA [35], which relies on sophisticated attention mechanisms, suggesting that our knowledge-centric reasoning is more effective for disambiguation in short-text scenarios. The comparison with Sui et al. [37], who implicitly model emotion associations,

TABLE 3. Performance comparison with SOTA models on the GoeMotions test set.

Model	Core Method/Contribution	Macro-F1 (%)	Micro-F1 (%)	Hamming Loss (↓)
BERT (Sitoula et al. [33])	Standard PLM fine-tuning	51.3	63.2	0.135
RoBERTa (Sitoula et al. [33])	Stronger PLM fine-tuning	54.1	65.9	0.121
EmoRoBERTa (López-López et al. [34])	Domain-adapted RoBERTa	54.9	66.5	0.118
EmoBERTa-X (Labib et al. [36])	RoBERTa + multi-head attention + ensemble	56.2	67.8	0.112
Singh et al. [39]	Multi-task learning with emotional definitions	57.1	68.3	0.109
HGCN-EC (Chen et al. [38])	Hierarchical knowledge + GCN	57.5	68.9	0.106
Sui et al. [37]	Transformer + masked emotion modeling	58.2	69.1	0.101
Ramakrishnan et al. [40]	BERT + simplified ASL	56.8	68.1	0.110
PK-GAT (Our Model)	Path-aware knowledge injection + GAT + ASL	58.72	69.85	0.098

TABLE 4. Performance comparison with SOTA models on SemEval-2018 E-C test set.

Model	Core Method/Contribution	Macro-F1 (%)	Micro-F1 (%)	Hamming Loss (↓)
RoBERTa-MA (Ameer et al. [35])	RoBERTa + multi-attention heads	60.1	64.5	0.151
Sui et al. [37]	Transformer + masked emotion modeling	61.3	65.2	0.145
BERT + Clipped ASL (Ramakrishnan et al. [40])	BERT + simplified ASL	61.8	65.7	0.142
PK-GAT (Our Model)	Path-aware knowledge injection + GAT + ASL	63.24	67.08	0.136

also indicates the strength of our explicit modeling with GAT and path reasoning.

Most notably, PK-GAT’s performance exceeds Ramakrishnan et al. [40], who similarly use ASL. This superiority can be attributed directly to our more complex, knowledge-aware architecture, confirming that combining a robust loss function with advanced knowledge injection and explicit label dependency modeling unleashes the full potential of multi-label emotion recognition.

2) EFFICIENCY COMPARISON WITH STATE-OF-THE-ART MODELS

Beyond superior predictive performance, the practical utility of an advanced model also hinges on computational efficiency and resource demands. To thoroughly assess the PK-GAT framework, we conducted a detailed quantitative analysis comparing model parameters, floating-point operations (FLOPs), and inference speed measured as frames per second (FPS) against selected baseline models. Parameters indicate model storage size, FLOPs represent computational complexity per forward pass, and FPS directly reflects inference speed on specific hardware.

All evaluations were conducted on a single NVIDIA Tesla V100 GPU, with batch size standardized at 32 and text sequence length padded to 128 tokens.

Several key insights emerge from the results in Table 5:

Firstly, compared to approaches employing large pretrained models (e.g., RoBERTa-large) or complex graph structures built upon them (e.g., HGCN-EC), PK-GAT demonstrates significant advantages in terms of both model size and computational cost. This advantage stems primarily from our decision to utilize DeBERTa-v3-base rather than its larger counterpart. Our carefully designed architecture, combined with efficient external knowledge integration, achieves comparable or superior performance without the need for extremely large models, making it particularly beneficial for deployment in resource-constrained scenarios.

Secondly, relative to Sui et al. [37], who also use a base-level pretrained model, PK-GAT shows slightly higher FLOPs per single forward pass. This additional computational overhead mainly arises from the path-aware knowledge injection module—especially path encoding and ranking—and the GAT. Notably, however, Sui et al.’s model employs an iterative decoding strategy during inference to achieve optimal performance, significantly impacting inference speed as reflected in FPS. In contrast, our PK-GAT model operates on a single-pass inference paradigm. Despite slightly higher per-pass computational cost, PK-GAT maintains a high inference speed at approximately 85 FPS, effectively balancing computational efficiency and predictive power.

TABLE 5. Efficiency and resource consumption comparison.

Model	Parameters (M)	FLOPs (G)	FPS (samples/sec)
RoBERTa-large (baseline)	355	90.2	110
HGCN-EC (Chen et al. [38])	360	93.5	102
Sui et al. [37] (with iterative decoding)	115	35.7	45 (iterative)
PK-GAT (our model)	135	48.6	85

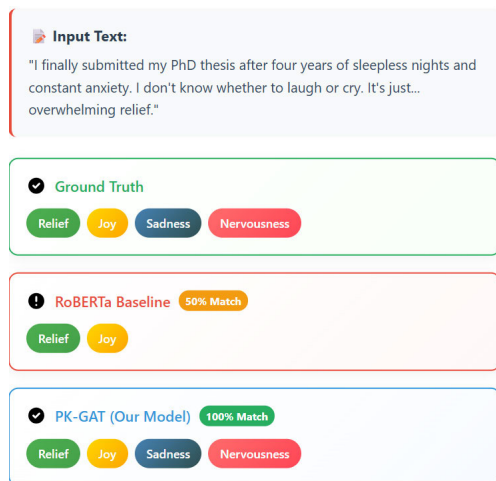
Collectively, the PK-GAT design philosophy focuses on maximizing model performance by integrating structured external knowledge and explicitly modeling label interdependencies, without substantially compromising computational efficiency. Experimental results from Tables 3-5 confirm that PK-GAT meets this goal, establishing state-of-the-art performance while ensuring practicality in real-world applications through manageable model size and fast inference speeds.

F. CASE ANALYSIS

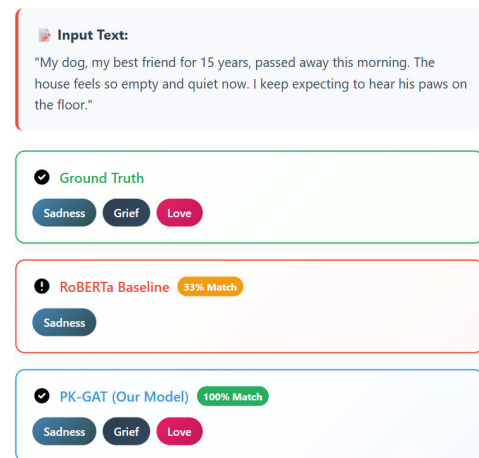
To gain a clearer view of how our PK-GAT framework operates and where it shines—or stumbles—we examined a set of real examples drawn from the GoEMotions test set. By looking at both successes and failures, we can trace the model’s reasoning when handling complex emotional expressions.

1) SUCCESSFUL CASES

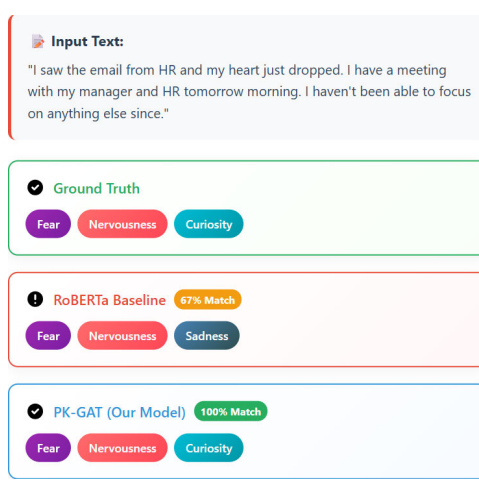
From Figure 7, RoBERTa correctly identifies Relief and Joy. In contrast, our PKI module then activates the path anxiety → manifests_as → nervousness, supplying clear evidence for Nervousness and Sadness. Meanwhile, the GAT layer leverages learned co-occurrence patterns—such as Relief often appearing alongside Sadness—to boost those lesser cues. The result is a full, precise emotion set.

**FIGURE 7.** Successful case 1: recognizing complex concurrent emotions.

From Figure 8, while the baseline picks up Sadness, the deeper emotions of Grief and Love require reasoning. PK-GAT’s advantage lies in path-aware reasoning. For Grief,

**FIGURE 8.** Successful case 2: reasoning with commonsense knowledge.

the model links “passed away” and “best friend” and finds a strong causal path in SenticNet: loss_of_loved_one → causes → grief. For Love, the phrase “best friend for 15 years” triggers a path like long-term_companionship → is_a → form_of_love. This lets the model infer Love as the root of Sadness and label it accordingly.

**FIGURE 9.** Successful case 3: distinguishing subtle negative emotions.

From Figure 9, the key here is to differentiate between Sadness and Curiosity. The baseline associates “heart just dropped” with Sadness, but PK-GAT identifies the core event

as an upcoming meeting—a source of uncertainty. In SenticNet, uncertainty often leads to both curiosity and fear. PK-GAT uses this to assign Curiosity, capturing not just anxiety but also the natural urge to know what will happen next.

2) FAILURE CASES

From Figure 10, this example highlights a classic failure to detect irony. The model takes praise and “unforgettable” literally, activating knowledge paths for positive sentiment, and misses the pragmatic reversal signaled by “Given that...”. It shows a gap in handling nonliteral meaning and deeper social context.

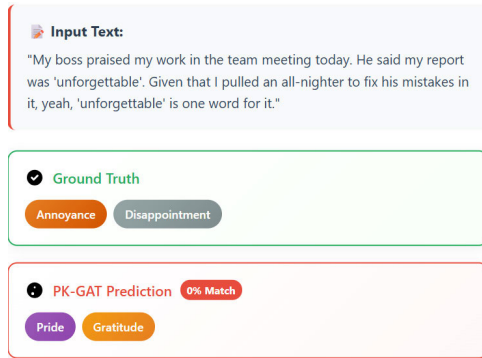


FIGURE 10. Failure case 1: challenges with cultural metaphor and irony.

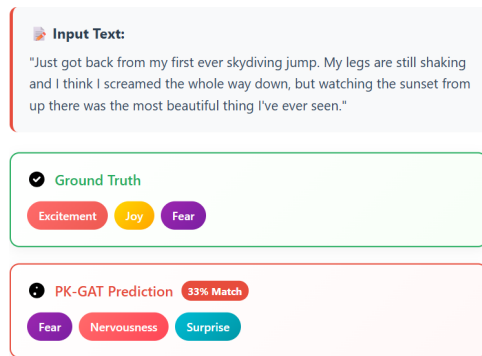


FIGURE 11. Failure case 2: confusing primary and secondary emotions in mixed states.

From Figure 11, the model correctly identifies the physical responses related to Fear and Nervousness but weighs them as primary emotions. It misses the emotional summary in the final line about beauty and awe, which points to Excitement and Joy. This suggests the model can track emotional signals but may struggle to recognize which are dominant and which are secondary, especially when narrative context shifts at the end.

From Figure 12, this is an error rooted in general commonsense. The model finds links like *watching_tv* → *causes* → *amusement/joy* in SenticNet, predicting Amusement, Joy, and Relief. However, “BoJack Horseman” is famous for its

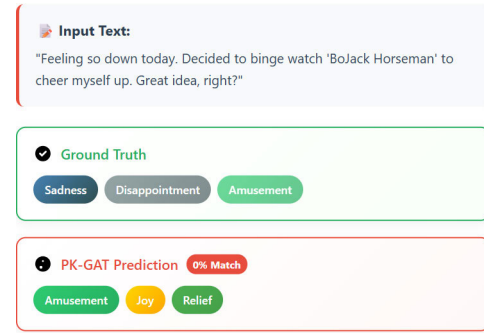


FIGURE 12. Failure case 3: missing or incorrect commonsense in the knowledge graph.

bleak exploration of sadness and nihilism. Watching it rarely brings joy—often the opposite. The self-mocking “Great idea, right?” is also missed. Since SenticNet and most general knowledge graphs lack culture-specific entries, PKI is misled, exposing limitations when facing context-dependent or pop-culture references.

V. CONCLUSION

In this work, we addressed the intricate challenge of fine-grained, multi-label emotion recognition, with a particular focus on its application to understanding complex psychological texts. We argued that a deeper level of understanding requires a model to move beyond surface-level text interpretation and to incorporate both commonsense reasoning and an awareness of inter-emotion dynamics. To this end, we introduced PK-GAT, a cohesive framework that synergistically combines three key ideas: a powerful deep contextual text encoder, a novel path-aware knowledge injection mechanism, and a dynamic graph-based label correlation model. Our empirical evaluation, conducted on the large-scale GoeMotions dataset and the widely-used SemEval-2018 benchmark, confirmed the effectiveness of our approach. PK-GAT consistently outperformed a wide array of strong baselines and recent state-of-the-art models, demonstrating significant gains, especially on the Macro-F1 metric, which underscores its ability to better recognize less frequent, more nuanced emotions. The comprehensive ablation studies further substantiated our architectural design choices. We showed that explicitly encoding and ranking multi-hop semantic paths from SenticNet provides a far richer signal than simply using node-level knowledge. Likewise, modeling the label space with a GAT proved crucial for capturing emotional co-occurrence and refinement.

However, our case analysis also transparently highlighted the current limitations of our work. Like most contemporary models, PK-GAT struggles with non-literal language, such as sarcasm and irony, and its reasoning is bounded by the knowledge contained within its external graph, making it vulnerable to missing context-specific cultural information. These challenges illuminate clear directions for future research. Promising avenues include integrating modules for

pragmatic understanding, such as sarcasm detection, and exploring methods for creating and maintaining dynamic, personalized knowledge graphs that evolve with user interactions. Further work could also investigate the fusion of multi-modal signals, such as vocal tone and facial expressions, to build a truly holistic model for emotion assessment in mental health contexts.

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